

Embedded Systems Team - 22

MULTI AGENT SMART BRICKS

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Faculty and Mentors

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Problem Statement

- Clinical Hazard: Retained Surgical Instruments pose severe risks to patient safety.
- Manual Error: Current counting relies on manual human checks under high stress & fatigue.
- **Solution: autonomous ESP32 'Smart Bricks' forming a modular, self-organizing tray.**
- Expected Outcome: Major Instrument accountability in real time with minimal failure.

Motivation

1. Domain Overview & Existing Work:

- Smart ORs explore RFID tool cabinets.

2. Existing Gaps:

- Attaching bulky sensors directly to tools hinders surgeon ergonomics & sterilization.
- Current protocol relies on nurses and surgical technicians performing manual verbal whiteboard counts before, during, and after long, high-stress surgeries.

3. Why Modular Bricks?

- Distributed peer-to-peer compute ensures no failure in terms of Instruments; high adaptability .

Proposed Solution

1. Plan of Action:

- Modular ESP32 bricks with custom instrument grooves acting as smart tray segments.

2. State-of-the-Art Comparison:

- Centralized Wi-Fi >>> Replaced by direct low-latency ESP-NOW radio mesh.
- Active tool electronics >>> Replaced by passive RFID tags seated in custom brick indents.

3. Key Features & Functionalities:

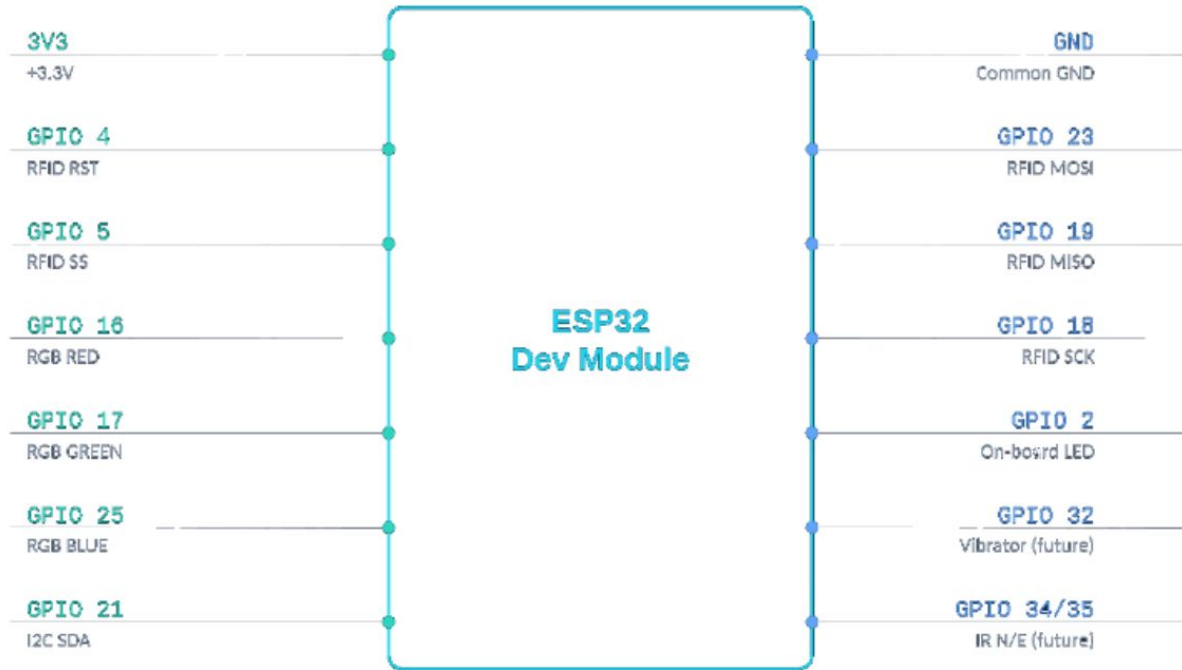
- Multi-Directional Neighbor Detection: Directional IR handshake on the Four directions.
- Decentralized Topology Solver: Reconstructs relative 2D grid coordinates automatically.
- Dynamic Visual Feedback: RGB LEDs transition colors based on active neighbor count.

Sensors Used & Diagrams

Hardware Architecture (Per Brick Cluster):

- Microcontrollers: ESP32 NodeMCU (Distributed peer-to-peer compute over ESP-NOW)
- Instrument Tracking: MFRC522 RFID Readers + Passive Tags
- Face Neighbor Detection: Directional IR Transceivers (4 faces/brick)
- Motion & Orientation: IMUs & Magnetometers (MPU9250) for heading tracking
- Actuation & Feedback: RGB LEDs, Active Buzzers, 6mm ERM Motors
- Enclosure: Modular 3D printed bases with instrument slots and translucent diffuser caps

Sensors Used & Diagrams



Initial Results

- 2-Brick Hardware Prototype Validated:
 - Direct ESP-NOW peer-to-peer packet broadcasting established.
 - RFID instrument presence tracking & timeout debouncing working reliably.
- Dynamic Neighbor LED Status Indication:
 - Red (0 Neighbors) | Yellow (1 Neighbor) | Blue (2 Neighbors) | Green (Full Tray)
- Multi-Agent Topology Simulation
 - Verified 2x2 grid, 1x4 linear strip, and dynamic disconnection aging.

Project Status

1. Current Demonstration Status:

- Standalone firmware operating autonomously .
- Topology engine, face mating logic, and neighbor tables fully implemented and verified.
- IR Handshake + RSSI (esp-NOW) Proximity consensus logic

2. Key Challenges & Mitigations:

- Optical IR Cross-Talk: Physical 3D directional channels and pulsed token sync.
- Power Distribution: Standalone LiPo battery support per modular brick.
- Heading Alignment: Closed-loop PID controller using magnetometer data and vibration motor.

Future Roadmap

IR Handshake Algorithm, IMU-Magnetometer Split, Haptics

- 4 photodiodes sensor data auto assembles the grid in a global adjacency list.
- 9 Axial Gyroscope (IMU-Magnetometer) To align coordinate frames
- Haptic ERM Motor with Tactile conformation
- Modular 3D printed enclosures

Milestones and Progress

